ProvenanceLab

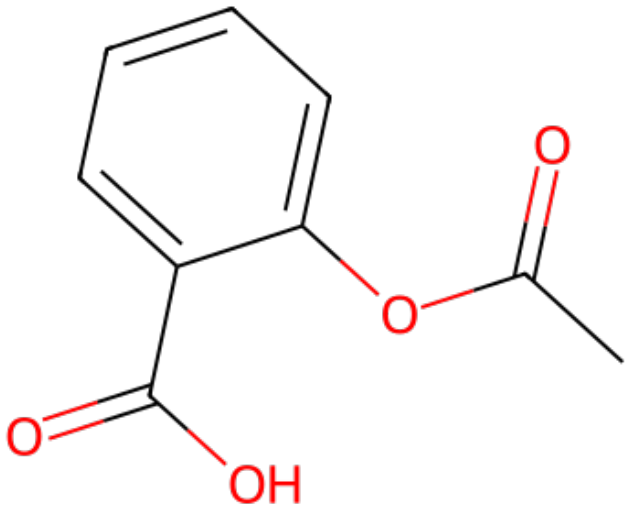
Deterministic molecular validation report

Research aid only. This report is not a prediction of efficacy or safety. Missing values are never inferred or silently filled.

Run ID	PLR-E6EE86C2BF-C8CF04
Structures	3
Input SHA-256	e6ee86c2bf1c7b27fa1650994db5d6737bc7a659d1c77cf313cfe1c4df3ae0cb
Configuration SHA-256	c8cf04efdb24dc7030612ff08fe9aaca1d0378fe4aa34c761ce44cc8baca776b
Output SHA-256	7de293c63d17d41d245494213b3f6eaef83a0380d2d5d7da3bd2a7053fc61e4c

1. Aspirin

Overall status: **VALID** | Analysis ID: PL-6AB2225D-C8CF



Submitted structure

`CC(=O)Oc1ccccc1C(=O)O`

Canonical and standardized forms

Canonical SMILES	`CC(=O)Oc1ccccc1C(=O)O`
Standardized parent	`CC(=O)Oc1ccccc1C(=O)O`

Validation findings

No validation issue was detected by the configured ruleset.

Standardization record

Step	Outcome	Before	After
Normalize functional groups and charges	No change	`CC(=O)Oc1ccccc1C(=O)O`	`CC(=O)Oc1ccccc1C(=O)O`
Select the parent fragment	No change	`CC(=O)Oc1ccccc1C(=O)O`	`CC(=O)Oc1ccccc1C(=O)O`
Normalize removable formal charges	No change	`CC(=O)Oc1ccccc1C(=O)O`	`CC(=O)Oc1ccccc1C(=O)O`
Select the canonical tautomer	No change	`CC(=O)Oc1ccccc1C(=O)O`	`CC(=O)Oc1ccccc1C(=O)O`

Property and evidence audit

Property	Computed	Evidence	Status / reason
Molecular formula	C9H8O4	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Molecular weight	180.16 g/mol	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Exact mass	180.0423 Da	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.

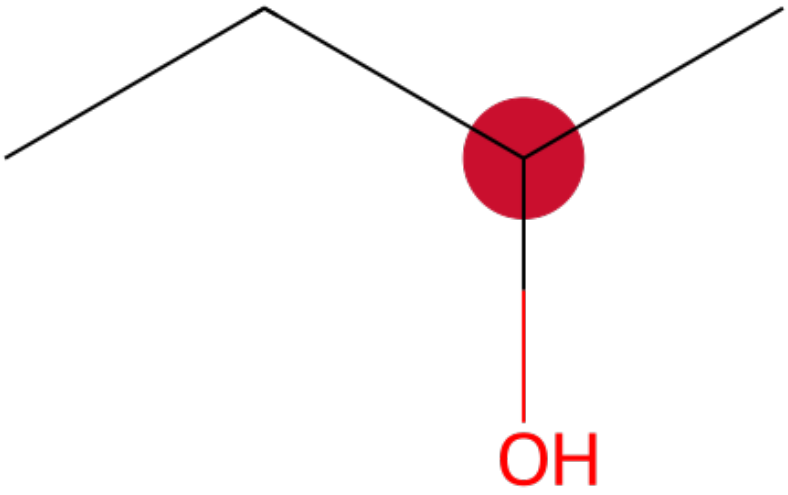
Property	Computed	Evidence	Status / reason
LogP	1.31 unitless	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Topological polar surface area	63.6 A^2	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Hydrogen-bond donors	1 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Hydrogen-bond acceptors	3 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Rotatable bonds	2 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Ring count	1 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Fraction Csp3	0.11 unitless	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.

Provenance

Input SHA-256	`6ab2225dbb47ed1354dd41abb3c7e3dbddcb9a52034fac8ee50487e445c4836a`
Configuration SHA-256	`c8cf04efdb24dc7030612ff08fe9aaca1d0378fe4aa34c761ce44cc8baca776b`
Output SHA-256	`63c4b1b764235095c254a4577ed07cd833cd196337e02d2d9f49a347d61284e2`
RDKit	`2026.09.1`
Ruleset	`hm-01.2-proposed`
Evidence snapshot	`not-configured`

2. 2-butanol

Overall status: **AMBIGUOUS** | Analysis ID: PL-38717C5A-C8CF



Submitted structure

`CCC(O)C`

Canonical and standardized forms

Canonical SMILES	`CCC(C)O`
Standardized parent	`CCC(C)O`

Validation findings

Code	Finding	Why it matters	Next step
`UNSPECIFIED_STEREOCHEMISTRY`	Potential stereochemistry at atom(s) 3 is not assigned.	Different stereoisomers can have different biological activity and properties.	Check the paper or source diagram for wedge bonds or an explicit stereochemical identifier.

Standardization record

Step	Outcome	Before	After
Normalize functional groups and charges	No change	`CCC(C)O`	`CCC(C)O`
Select the parent fragment	No change	`CCC(C)O`	`CCC(C)O`
Normalize removable formal charges	No change	`CCC(C)O`	`CCC(C)O`
Select the canonical tautomer	No change	`CCC(C)O`	`CCC(C)O`

Property and evidence audit

Property	Computed	Evidence	Status / reason
Molecular formula	C4H10O	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Molecular weight	74.12 g/mol	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.

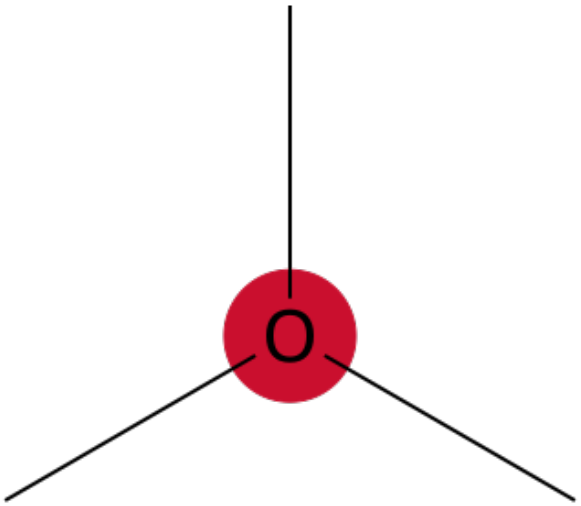
Property	Computed	Evidence	Status / reason
Exact mass	74.0732 Da	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
LogP	0.78 unitless	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Topological polar surface area	20.23 A^2	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Hydrogen-bond donors	1 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Hydrogen-bond acceptors	1 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Rotatable bonds	1 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Ring count	0 count	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.
Fraction Csp3	1.0 unitless	Unknown - see reason	**NOT_AVAILABLE** No frozen PubChem or ChEMBL snapshot is configured in this build, so a reference value cannot be reported.

Provenance

Input SHA-256	`38717c5af748ccf5d107586f4269c055bdaa8d49e0c52f6b782a78a35808f92e`
Configuration SHA-256	`c8cf04efdb24dc7030612ff08fe9aaca1d0378fe4aa34c761ce44cc8baca776b`
Output SHA-256	`fd6d1549c1158e28cd5cb059400dcf396c488924552f025f947458ec9462c3af`
RDKit	`2026.09.1`
Ruleset	`hm-01.2-proposed`
Evidence snapshot	`not-configured`

3. Invalid oxygen

Overall status: **INVALID** | Analysis ID: `PL-93B54212-C8CF`



Submitted structure

`C[O](C)C`

Canonical and standardized forms

Canonical SMILES	`Unknown - see reason`
Standardized parent	`Unknown - see reason`

Validation findings

Code	Finding	Why it matters	Next step
`IMPOSSIBLE_VALENCE`	The graph contains an atom with an incompatible combination of bonds and formal charge.	The structure is not chemically valid and cannot be used for safe property calculations.	Review the highlighted atom, formal charge, and surrounding bond orders in the source.

Standardization record

Not run because no valid molecular graph was available.

Property and evidence audit

Property	Computed	Evidence	Status / reason
Molecular formula	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Molecular weight	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Exact mass	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
LogP	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Topological polar surface area	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Hydrogen-bond donors	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.

Property	Computed	Evidence	Status / reason
Hydrogen-bond acceptors	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Rotatable bonds	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Ring count	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.
Fraction Csp3	Unknown - see reason	Unknown - see reason	**INVALID_INPUT** Evidence lookup is not applicable until the structure is valid.

Provenance

Input SHA-256	`93b54212e20151b6232482a287d8d0211de97c93002c4e1ec148ff976392e0f7`
Configuration SHA-256	`c8cf04efdb24dc7030612ff08fe9aaca1d0378fe4aa34c761ce44cc8baca776b`
Output SHA-256	`723ede03e5fbeb6e37544b62fb617ce7938c0ec06536589aa6738cc64f4c1cda`
RDKit	`2026.09.1`
Ruleset	`hm-01.2-proposed`
Evidence snapshot	`not-configured`